# Linux-2

Anything you type comes from the keyboard and is part of an input stream called stdin. Anything sent to the screen is sent first to the outstream or the stdout.

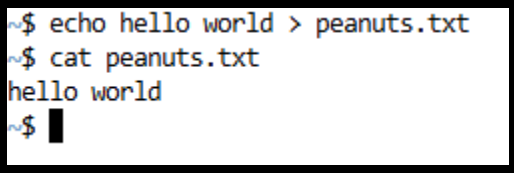
Linux normally takes the data in stdin and sends it to stdout. You can however redirect this using the redirection operator **>**.

Try the following command:

**echo Hello World > peanuts.txt**

Open the contents on the screen using the **cat** command:
**cat peanuts.txt**

Take a screenshot of the results:




In this command what was normally sent to the screen is now **redirected** to the file **peanuts.txt**. As a result a file called **peanuts.txt** will contain the **Hello Word** phrase rather than the screen or stdout.

**>>** redirects to a file without overwriting it.

Try the following:

**echo Hello World Again >> peanuts.txt**

Display the contents of the file again using the cat command:

**cat peanuts.txt**

Include a screenshot of the results:

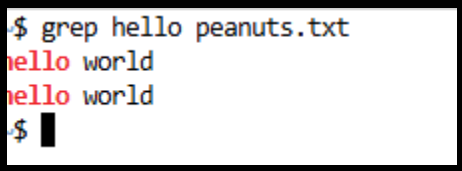
```
~$ echo hello world >> peanuts.txt
~$ cat peanuts.txt
hello world
hello world
~$
```

If you want to find text in a file use the grep command:
**grep fox sample.txt**

This looks for the word fox in the sample.txt file. Let's look for the word Hello in the peanuts.txt file as follows:

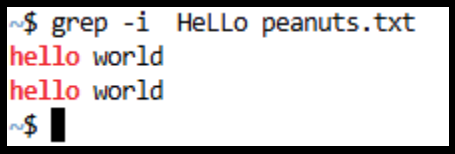
**grep Hello peanuts.txt**

Include a screenshot here:




Add the -i flag to make the search case insensitive. Try the following:
**grep -i hello peanuts.txt**

Include a screenshot here:




You can even pass the stdout of a command to grep to search for strings in it using the **pipe operator** |

Try using the command env:
**env**

Include a screenshot:




You should notice your screen is filled with a large amount of data. We can call this command and pass the output (stdout) to another command using the pipe operator as follows:

**env | grep -i User**

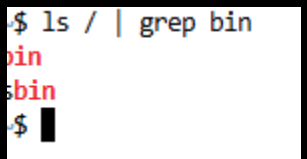
Try this and take a screenshot:




This passes the stdout of the env command to grep which looks for the word User without worrying about case. The | is called the pipe operator. It allows us to pass the stdout of a command to the stdin of another.

Here's another example:
**ls / | grep bin**

Take a screenshot and embed it here:



Explain how this works:

The ls command lists the contents of the directory, grep bin finds the word “bin” in the directory

**Text Editors**

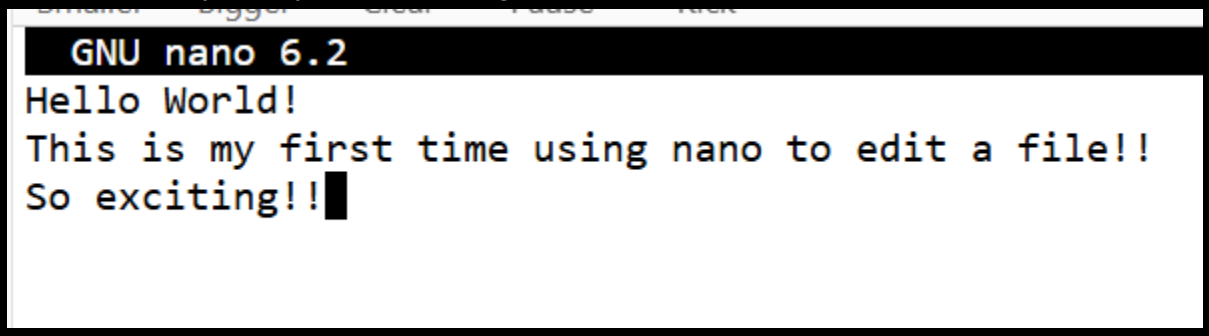
The Linux shell comes with a variety of different text editors to allow you to open, edit and save changes to different files. The one we will use is called nano. Type the following:

**nano file1.txt**

You should see the following:

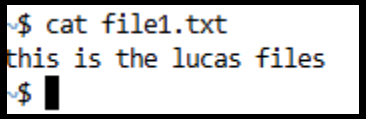



This is an empty file. Type the following in it:




To save it press the following key combinations:
**CTRL + X**
**Y**
**Enter**
You should be back in the terminal. Type **ls** to see the contents of your current directory. You should see the **file1.txt** there.

Display the contents of the file using the **cat** command. Include a screenshot here:



**Users and Groups**

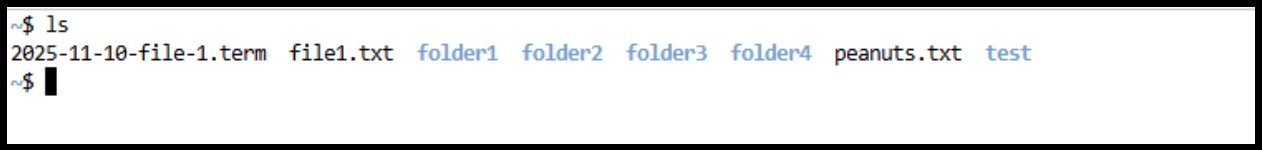
There can be many users in a single computer running Linux. Specific users only have access to their own files unless they are part of a group that has been assigned certain privileges to certain files as well. Users can run their own processes that are separate from other users. The most powerful user is the superuser or sudo or sometimes referred to as the root user.

If you encounter an error with permissions then you will have to be a superuser and precede your command with sudo. You'll be asked to provide your password.

Try the following:

**mkdir /test**

Include a screenshot of the results and explain it:




It makes directory

Now try the following:

**sudo mkdir /test**

Sudo doesnt work on cocalc

List the contents of the root directory and you should see the folder test there. You cannot do whatever you want wherever you want in a linux machine unless you're a superuser. The only place you can do almost anything without superuser privileges is your personal folder in the /home directory.

Questions and Exercises:

1. Make sure you are connected to the internet (get a wifi adapter). Let's use the Debian package manager to download and play some games.
   **sudo apt install bastet**

   To play it just type:
   **bastet**

   Here's another:
   **sudo apt install ninvaders**

To play it just type: **ninvaders**

Why did we have to download and install these games using sudo?

Because you need superuser permission to install

2. Create a file called "**poems.dat**" with a short poem using echo and the redirection operator. The file should be named **poem.dat**.

3. Using the file append >> operator add a last line that includes your full name.

4. Make a copy of the "**poems.dat**" file by using the cat command and redirecting its output to a file called **poems1.dat**.

5. Using the cat command, display the contents of each file.

6. Use the following command to download a data file (use CTRL+C in the GUI to copy and CTRL+SHIFT+V to paste into the terminal):

   **wget https://www.stats.govt.nz/assets/Uploads/Annual-enterprise-survey/Annual-enterprise-survey-2023-financial-year-provisional/Download-data/annual-enterprise-survey-2023-financial-year-provisional.csv**

7. List the current folder contents using **ls**. Find the file and make a copy with a simpler name i.e. **data.txt** by using the **cp** command:

8. Use **cat** to display the contents of **data.txt**.

9. Let's download a repository called mystery:
   **wget https://github.com/veltman/clmystery/archive/refs/heads/master.zip**

   Next we need to unzip this compressed file (check the name of the downloaded zip file):
   **unzip master.zip**

10. Use **ls** to display the contents of your current directory. What do you notice is there now that wasn't there before?

11. Enter the directory **clmystery-master** using the **cd** command.

12. List the contents of the new directory. Open and read the **README.md** file using the cat command.

13. Now read the instructions file.

14. Enter the **mystery** folder.

15. Read the **crimescene** file that is in the **mystery** folder.

16. It's a huge file with too much to read and so little time. Fortunately there is the grep command which helps us filter data. Use the command
    **grep "CLUE" crimescene**

    This will filter those lines that have the word "CLUE" in it.

17. One clue mentions a **female** left the **bar** and her name was probably **Annabel**.

18. Let's use the grep command on the people file to find everyone with that name:
    **grep "Annabel" people**

19. Your results show two females with the name Annabel: **Annabel Sun** and **Annabel Church**. Sun lives at 26 Hart Place, line 40 while Church lives at 38 Buckingham Place, line 179. These are our two possible witnesses. Let's interview them by looking at the streets directory and using the address information we just collected.

20. Type in the following while in the streets directory:
    **head -n 40 Hart_Place | tail -n 1**

    We get the number 40 from the line number associated with the address. The head command will read in a file and by using the -n flag followed by a number it displays the first 40 lines of the file. We then pass these 40 lines via the pipe | operator into the tail command and use the -n flag again to read the last line only (the tail command reads a set number at the end of a list of contents).
    You will see the following:
    SEE INTERVIEW #47246024

21. Now do the same for the other witness:

**head -n 179 Buckingham_Place | tail -n 1**

We get the following:
SEE INTERVIEW #699607

22. Read the interview file as follows (go to the interviews folder):
    **cat interview-47246024**

23. From this you will find that Ms. Sun is not our witness as she doesn't fit the description given by the Barista. That means Ms. Church is our key witness.

24. Read the interview file as follows (go to the interviews folder):
    **cat interview-699607**

25. In this interview Ms. Church states that she saw a blue Honda with a license plate that starts with "L337" and ends with "9" flee the scene.

26. Let's find who owns this car by typing (go to the mystery folder where the vehicles folder resides):
    **grep -A 7 "L337" vehicles**
    This will find and display the find and display the 7 lines after each match.

You will find at least two men at least 6’ feet tall with a blue honda match the license plate description: 1. Jeremy Bowers and 2. Joe Germuska

27. Let’s go back to our clues: One of them states that the killer had memberships to AAA, Delta SkyMiles, the local Library and the Museum of Bash History. Let’s see
    Go to the memberships folder.
    if our two suspects are members in any of these:
    **grep “Joe” AAA**

    This shows that Joe belongs to the AAA.
    **grep “Jeremy” AAA**

    So is Jeremy.

    **grep “Jeremy” Delta***
    When checking in Delta_SkyMiles only Jeremy is a member!

    Jeremy is also the only member of the library!
    **grep “Jeremy” Terminal***

    It looks like we have found our killer!

28. Read the solution file to see if you are correct.

we  yes WE were correct

Jeremy going to jail
To become frail
He must be buggin
HE MUST NOT KNOW THAT IM THUGGIN